

Theory Questions :-

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- Q1. Explain quadratic time complexity i.e. $O(n^2)$.
- Q2. Explain logarithmic complexity like $O(\log n)$ or $O(n \log n)$.
- Q3. What is $O(2^n)$ – Exponential Time?
- Q4. What is $O(n!)$ – Factorial Time?
- Q5. Explain the priority/order of time complexity.
- Q6. Explain space complexity and how we can calculate it.
- Q7. What is the best case, average case, and worst case time complexity?
- Q8. What is constant space complexity?
- Q9. What is linear space complexity?
- Q10. What is quadratic space complexity?
- Q11. What is auxiliary space?
- Q12. How would you define a function?
- Q13. What is the difference between a function and a method?
- Q14. Can you explain the difference between function declaration and function definition?
- Q15. What is the difference between call by value and call by reference?
- Q16. What are pure functions? Can you give an example?
- Q17. What do you mean by side effects in functions?
- Q18. What is function overloading? How is it implemented in Java/C++?

MCQ Questions :-

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Q1. What will be the output?

```
for(int i=1; i<=3; i++){  
    for(int j=1; j<=2; j++){  
        System.out.print(i + "" + j + " ");  
    }  
}
```

- A) 11 12 21 22 31 32
- B) 12 21 31
- C) 11 22 33
- D) Compilation Error

Q2. Which statement is true about functions in Java?

- A) Function can return multiple values directly
- B) Function must always return value
- C) Function can be called many times
- D) Function cannot take parameters

Q3. What will be the output?

```
static int fun(int n){  
    if(n==0)  
        return 0;  
    return n + fun(n-1);  
}
```

```
System.out.println(fun(3));
```

- A) 3
- B) 6

- C) 5
- D) 0

Q4. What is recursion?

- A) Calling one class from another class
- B) Calling function inside loop
- C) Function calling itself
- D) Creating object repeatedly

Q5. What will be the output?

```
static void show(int n){  
    if(n==0)  
        return;  
    System.out.print(n + " ");  
    show(n-1);  
}
```

show(3);

- A) 1 2 3
- B) 3 2 1
- C) 0 1 2
- D) Error

Practice Questions :-

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Q1.Problem Statement:

Write a Java program to create a function that accepts one integer number and prints its multiplication table up to 10.

Explanation:

Create method table(int n) and use loop inside function.

Input:

5

Output:

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

...

5 x 10 = 50

Q2.Problem Statement:

Write a Java program to create a recursive function that returns sum of first N natural numbers.

Explanation:

Use formula through recursion:

sum(n) = n + sum(n-1)

Input:

5

Output:

15

Q3.Problem Statement:

Write a Java program to create a function that accepts a number and checks whether it is prime or not.

Explanation:

Prime number has exactly two factors: 1 and itself.

Input:

7

Output:

Prime Number

Q4.Problem Statement:

Write a Java program to print first N Fibonacci numbers using recursive function.

Explanation:

Series: 0 1 1 2 3 5 8 ...

Input:

6

Output:

0 1 1 2 3 5

Q5.Problem Statement:

Write a Java program to create a function that accepts a number and returns total number of digits.

Explanation:

Use loop inside function and return count.

Input:

45891

Output:

5